SPRITLE

spatia

SPATIAL INTELLIGENCE FOR ROBOTS

SPATIA – PHYSICAL AI & ROBOTICS - Internship

About Spatia

Spatia.sg is the Physical AI and Robotics solutions division of Spritle Software, focused on bringing AI-powered automation to real-world industrial environments. We work with robotic OEMs, system integrators, and manufacturers on applications such as **bin picking, pick-and-place, assembly, and intelligent robot manipulation**.

Position: Physical AI & Robotics Internships
Duration: 6 Months
Location: Chennai
Employment Type: Internship

Roles :

1. Computer Vision, 3D Perception & VLM
2. Robotics & Manipulation

Who We're Looking For :

We value candidates with **strong fundamentals, problem-solving ability, curiosity, and a willingness to learn**. Academic, personal, research, hackathon, or open-source projects in AI, Computer Vision, 3D Perception, Robotics, or related areas are highly relevant.

Strong fundamentals and the ability to learn and experiment are important.

What You'll Gain

- Hands-on experience with **real-world Physical AI and robotic systems**.
- Exposure to 2D/3D Computer Vision, VLMs, robotics, sensors, and AI models.
- Experience working with physical robotic arms, cameras, perception systems, and manipulation workflows.
- Opportunity to collaborate with AI, Vision, Robotics, Software, and Hardware teams.
- Experience in real-world robotics datasets, experimentation, evaluation, calibration, and deployment.

SPRITLE

spatia
SPATIAL INTELLIGENCE FOR ROBOTS

1. Computer Vision, 3D Perception & VLM Intern

What You'll Work On

- Develop 2D computer vision pipelines for object detection, classification, segmentation, tracking, and image analysis.
- Work with stereo, RGB-D, and depth cameras for 3D perception and scene understanding.
- Process 3D point clouds using Open3D and Python-based tools.
- Work on camera calibration, depth processing, coordinate transformations, and sensor-to-robot calibration.
- Develop perception pipelines using OpenCV, NumPy, SciPy, PyTorch/TorchVision.
- Experiment with Vision-Language Models (VLMs) for image understanding, visual reasoning, object identification, and task interpretation.
- Explore Vision-Language-Action Models (VLAMs), foundation models, and embodied AI for robotic applications.
- Build datasets, prepare annotations, evaluate models, and perform inference/benchmarking.
- Integrate vision and AI pipelines with robotic applications and real-world sensors.

Must Have

- Bachelor's/Master's degree or final-year student in Computer Science, AI/ML, Robotics, Mechatronics, Electronics, or related field.
- Strong fundamentals in Python.
- Basic-to-good understanding of Computer Vision and image processing.
- Hands-on exposure to OpenCV, NumPy, and PyTorch/TorchVision.
- Understanding of basic 2D/3D geometry, depth, point clouds, coordinate frames, and transformations.
- Familiarity with Linux/Ubuntu and Git.

Good to Have

- Open3D and 3D point-cloud processing.
- RGB-D/stereo cameras such as Orbbec, RealSense, or equivalent.
- YOLO, Grounding DINO, SAM/SAM2, or equivalent models.
- VLMs such as Qwen-VL or similar multimodal models.
- ROS2, robotics simulation, Docker, CUDA, or NVIDIA platforms.
- Camera/hand-eye calibration or exposure to robotics and embodied AI.

2. Robotics & Manipulation Intern

What You'll Work On

- Develop and test applications for 6-DoF/7-DoF robotic manipulators such as Flexiv, Universal Robots, LeRobot, or equivalent platforms.
- Work with joint space, Cartesian space, end-effector poses, coordinate frames, and transformations.
- Implement and test Forward Kinematics (FK) and Inverse Kinematics (IK).
- Develop motion and trajectory execution workflows.
- Work with ROS2 nodes, topics, publishers, subscribers, services, TF/TF2, and robot interfaces.
- Use MoveIt2 or equivalent frameworks for motion planning and collision checking.
- Integrate robotic manipulators with cameras and perception systems.
- Perform robot, camera-to-robot, and hand-eye calibration.
- Develop pick-and-place, manipulation, and automation experiments on physical robots.
- Test robotic applications in simulation and transfer validated workflows to physical robots.

Must Have

- Bachelor's/Master's degree or final-year student in Robotics, Mechatronics, Mechanical Engineering, Electronics, Computer Science, or related field.
- Strong understanding of robotics fundamentals.
- Understanding of 6-DoF robotic arms and basic manipulator concepts.
- Good understanding of 3D Cartesian space, coordinate frames, and transformations.
- Understanding of Forward and Inverse Kinematics concepts.
- Programming experience in Python; C++ exposure is an advantage.
- Familiarity with Linux/Ubuntu and Git.

Good to Have

- ROS2 and TF/TF2.
- MoveIt2, OMPL, or other motion-planning frameworks.
- Flexiv, Universal Robots, LeRobot, or similar robotic arms.
- Robotics simulators such as Gazebo, Isaac Sim, or MuJoCo.
- Jacobians, singularities, trajectory generation, collision detection, and avoidance.
- Sim-to-real workflows.
- Robotic grippers/end-effectors.
- Vision-guided manipulation, 3D perception, VLM/VLAM, or AI-based robotic manipulation.